Kaylie Hynes

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EDUCATION

University of St Andrews – School of Computer Science *St Andrews, Scotland*

Master of Computer Science

Concentration*: *Computer Science

- **Modules:** Product and Project Management, Masters Programming Projects, Machine Learning for Data Analysis, Software Architecture & Design
- **Current Academic Research:** Dissertation focused on Deep Learning–driven disease detection in MRI imaging, exploring Convolutional Neural Networks and algorithmic methods to support earlier and more accurate identification of brain pathology.

Pace University – Ivan G. Seidenberg School of Computer Science and Information Systems *Financial District, NYC*

Bachelor of Science

***Major**: Information Technology; Concentration in Software Development*

- **Honors & Advanced**: Pace Incentive Award, Pace Institutional Grant Recipient, Bloomberg Market Concepts Certification

PROJECTS

Biomedical Research Experience: *MSc Dissertation - Deep Learning for Glioma MRI Analysis* *University of St Andrews*

- Developed deep learning pipelines for automated glioma segmentation using the BraTS 2021 and 2024 MRI datasets.
- Implemented and evaluated both 2D and 3D U-Net convolutional neural network architectures for brain tumor segmentation.
- Investigated model generalization across datasets, pre-treatment versus post-treatment imaging, and explainable AI methods to improve clinical interpretability.
- Conducted a comprehensive literature review to identify gaps in current brain tumor segmentation approaches and designed experiments to address them.
- Utilized Python, PyTorch, MONAI, NiBabel, and medical image analysis techniques to develop and evaluate models.

Research Assistant – Dr Allodi Laboratory
School of Neuroscience *University of St Andrews*

- Developing a 3D image analysis pipeline to quantify pre- and postsynaptic glycinergic inputs in light-sheet microscopy data from an hTDP-43 mouse model of ALS.
- Extending previous 2D image analysis methods to full volumetric analysis, addressing an identified gap in the laboratory's existing research.
- Designing computational workflows to process and quantify complex biological imaging datasets.
- Collaborating with neuroscience researchers to extract biologically meaningful measurements from high-resolution microscopy data.

WORK & LEADERSHIP EXPERIENCE

Dr Frank Scafuri & Associates **Staten Island, NY**
Medical Office Operator *2020-2025*

- Audit and organize patient records across multiple practitioners to identify outstanding referrals, follow-ups, and workflow gaps.
- Designed and maintained Trello-based tracking workflows to organize patient follow-ups, referral status, and outstanding actions.
- Analyze large volumes of patient chart information to identify missing follow-up actions and improve information organization.
- Coordinate information between practitioners, patients, and external healthcare providers through phone and administrative workflows.
- Manage and maintain sensitive healthcare information with a focus on data accuracy, organization, and confidentiality.
- Identify opportunities to improve administrative workflows and streamline tracking of patient-related tasks.

Law Office of Helene Hartig **233 East 86th St, New York, NY**
Paralegal *2024-2025*

- Leveraged legal information systems including LexisNexis, eCourts, and NYSCEF to conduct digital research, manage filings, and maintain structured matter data.
- Supported attorneys through technology-enabled document workflows, calendaring systems, and automated tracking of litigation events.
- Designed repeatable processes for intake, communications logging, and document preparation, increasing operational consistency.
- Maintained high-volume multi-channel communications (calls, emails, filings) with rigorous data governance practices

KI Legal **40 Wall Street, New York, NY**
Data Clerk *2023-2024*

- Executed large-scale data cleansing, normalization, and archival operations to maintain integrity across enterprise document repositories.
- Built and deployed custom applications and workflow automations within NetDocuments, accelerating retrieval, routing, and compliance processes
- Applied technical expertise to improve data management and automate tasks.
- Identified inefficiencies in legacy procedures and implemented system improvements that enhanced team productivity and reduced manual handling.

Appellate Associates **John Street, New York, NY**
Legal Data Entry Clerk (Temp) *2023*

- Maintained accuracy and consistency across case management databases through structured data ingestion and verification
- Collaborated with cross-functional teams to reconcile information discrepancies and uphold evidentiary and procedural standards.
- Contributed to digital file organization initiatives that improved accessibility and audit readiness

Gusrae Kaplan Nusbaum, LLC **New York, NY**
Billing Supervisor (Intern) *2022*

- Administered digital billing datasets, validating charges, tracking expenses, and ensuring compliance with firm policies
- Produced and distributed invoices through technology platforms while maintaining precise financial records
- Balanced academic commitments with responsibility for time-sensitive financial operations in a professional services environment.

VOLUNTEER EXPERIENCE

Pace University Criminal Justice Club

Secretary

- Responsible for the administration of the club
- Arranging meetings with just the club members and outside sources (ex. Police officers or different attorneys)
- Also responsible for the administration within the club and its members.

Pace University Data Science Club

Public Relations

- Organizing and attending promotional events for the club
- Managed outreach logistics, speaker coordination, and cross-team planning with creative and media groups to support technical programming.

ADDITIONAL SKILLS & INTERESTS

Technical Skill/Areas: Python, HTML, JavaScript, TensorFlow, Keras, Tkinter, Flask, Pandas, Web APIs, Microsoft Office Suite, Trello, NetDocuments, eCourts, NYSCEF, Data Management, Workflow Optimization, Information Systems, Documentation Control. IBDP High School Diploma Recipient, MATLAB, MONAI, Pytorch, MRI Analysis, NiBabel, scikit-learn, Matplotlib, CNNs, Deep Learning.

Interests: Software Engineering, Machine Learning, Data Analytics, Technology Applications in Healthcare and Law, LLMs, Software Development